

IMPLEMENTATION BY ORGANIZATIONS



THIS COURSE SO FAR: GROUP MEMBERS NEED LEADERS

Group members need leaders

They give them power to solve group's problems

They have tools to align leaders' behavior with their interests

- ▶ The threat of being fired
- ▶ Institutions constraining leaders' discretion

LEADERS SET POLICY GOALS

THE ORGANIZATION IMPLEMENTS THEM

Leaders need others to help them **implement** their goals

They need an **organization**

- ▶ Bureaucracy
- ▶ Employees
- ▶ Voluntary group members

Managing organizations is an essential skill of leaders

1. How to elicit effort?
2. How to attract “best” members to the organization
3. How to get information from your members?

HOW TO STRUCTURE INCENTIVES IS A PRIMARY PROBLEM FACING ORGANIZATIONS

This week we focus on how to ensure that organization members exert **effort** when implementing leader's decisions

In this course, **effort** denotes any action that is

- ▶ Desired by the leader
- ▶ Costly for the organization member

For example

- ▶ Physical or mental effort
- ▶ Pursuing leader's and not member's own objectives
- ▶ Not diverting resources from the organization

GROUP DISCUSSION

Form groups of 3-4 people

Why do people exert effort in organizations?

INCENTIVIZING EFFORT OF LEADERS VS OF GROUP MEMBERS

Group members who incentivize leaders

- ▶ Have limited tools—standard contracts not available
- ▶ Are heterogenous and have conflicting objectives

Leaders who incentivize group members

- ▶ Have more tools—including monetary rewards
- ▶ Are making decisions single-handedly

OUTLINE

INTRINSIC MOTIVATION AND CAREER CONCERNS

MONETARY INCENTIVES

INTRINSIC MOTIVATION AND PERFORMANCE PAY

INTRINSIC MOTIVATION

Intrinsic motivation: a person exerts effort because the task is *inherently satisfying*

- ▶ Purpose
- ▶ Mastery
- ▶ Autonomy
- ▶ Identity
- ▶ Prosocial impact

But may lead to effort that is not desired by the leader

- ▶ Beneficial only if person's and leader's objectives align

Mission alignment: the degree to which agents internalize organizational goals

CAREER CONCERNS

Career concerns: incentives to exert effort to influence outsiders' perception of person's ability or behavior, aimed at maximizing future benefits outside of her current position

- ▶ Next job
- ▶ History books

Can be a powerful motivator

But may lead to effort that is not desired by the leader

- ▶ Beneficial only if person's and leader's objectives align

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MAIN IDEA

People do things they are rewarded for!

To elicit effort, pay people more for more effort

SIMPLE SOLUTION

Tell person how much effort you want from them

Write a contract that promises them

- ▶ Employment at wage w if they exert that amount of effort
- ▶ Fire them otherwise

Problem: You do not observe effort!

Possible solution: Find a measure that is correlated with effort and reward them base on that

- ▶ Reward sales not effort
- ▶ Reward students' scores not effort

MORAL HAZARD

Moral hazard is the problem that the informed party might take actions that the other party cannot monitor and that takes advantage of the less-informed party.

In a principal-agent problem, the agent's effort is not observable

EXAMPLE OF THE MAIN PROBLEM

Organization: School district

Goal: Well educated children

Worker: Teacher hired to achieve the goal

Effort: Educating requires effort

Problem: Teacher's effort neither observable nor verifiable

Performance: Student scores (effort + luck)

EXAMPLE OF THE MAIN PROBLEM

Organization: School district **Principal**

Goal: Well educated children

Worker: Teacher hired to achieve the goal **Agent**

Effort: Educating requires effort **Effort**

Problem: Teacher's effort neither observable nor verifiable

Performance: Student scores (effort + luck) **Output**

BASIC COMPENSATION SCHEMES

Retention rule

- ▶ Decision to retain the agent or fire him (what we have been doing so far!)

Fixed salary

- ▶ Fixed salary independent of agent's performance

Performance pay

- ▶ Base pay plus bonus tied to performance

Should principal offer performance pay to incentivize effort?

MODEL OF PERFORMANCE PAY

A principal considers hiring an agent to produce output

Output (y) depends on agent's effort (e) and luck ($luck$)

$$y = e + luck$$

- ▶ Agent can exert no effort ($e = 0$) or exert effort ($e = 1$)
- ▶ Effort is costly: $c(e) = \alpha e$, with $\alpha < 1$
- ▶ Luck is random and has zero mean: $\mathbb{E}(luck) = 0$

MODEL: AVAILABLE CONTRACTS

Principal offers a performance-pay contract:

$$w = s + by$$

- ▶ s is the base salary
- ▶ Fixed wage is equivalent to $b = 0$
- ▶ $b > 0$ means pay for performance

Principal decides what s and b to offer

When pay is highly sensitive to performance (high b) we say incentives are **high powered**

TIMING

1. Principal offers contract specifying s and b
2. Agent decides if to accept
3. If contract signed, agent chooses effort e
4. Random events *luck* occur
5. Principal and Agent observe output y
6. Agent gets reward specified by contract

PAYOFFS

Agent:

Receives (outside option) $u < 1$ if she does not sign the contract

Otherwise agent choosing effort e expects to receive:

$$\mathbb{E}(w(e)) - c(e) = \mathbb{E}(s + b(e + \textit{luck})) - \alpha e = s + be - \alpha e$$

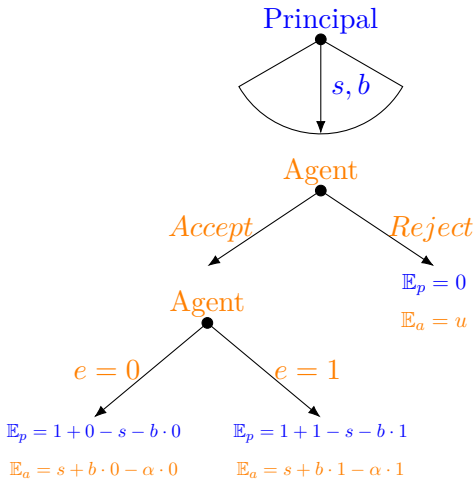
Principal:

Receives 0 if agent does not sign contract

Receives payoff of 1 from having an agent plus profit from production minus wage she pays the agent

$$1 + \mathbb{E}(y - w) = 1 + e - s - be$$

GAME TREE



WE SOLVE USING BACKWARD INDUCTION

Last stage:

Payoff from $e = 1$ is larger than payoff from $e = 0$ if :

$$s + b - \alpha \geq s$$

Agent chooses $e^* = 1$ if and only if $b \geq \alpha$

Performance pay elicits effort if bonus steep enough

WHEN WILL WORKER ACCEPT THE CONTRACT?

Second stage:

If $b < \alpha$, then $e^* = 0$, so Agent's payoff from the contract is s

- ▶ Agent accepts contract if $s \geq u$

If $b \geq \alpha$, then $e^* = 1$, so Agent's payoff from the contract is

$$s + b - \alpha$$

- ▶ Agent accepts contract if $s + b - \alpha \geq u$

First stage: Recall Principal chooses s and b to maximize

$$1 + \mathbb{E}(y - w) = 1 + e - s - be$$

subject to Agent accepting the contract

Principal will offer either

$$b^* = 0, \quad s^* = u$$

or

$$b^* = \alpha, \quad s^* = u - b + \alpha$$

WHAT CONTRACT WOULD PRINCIPAL OFFER?

Recall Principal's profit:

$$1 + \mathbb{E}(y - w) = 1 + e - s - be$$

Principal's profit from

$$b^* = 0, \quad s^* = u \quad \rightarrow \quad 1 - u$$

Principal's profit from

$$b^* = \alpha, \quad s^* = u - b + \alpha \quad \rightarrow \quad 2 - u - \alpha$$

Note that since $\alpha < 1$, we have

$$1 - u < 2 - u - \alpha$$

Principal offers performance pay

Optimal bonus depends on marginal cost of effort α

Optimal base salary depends on agent's outside option u

TAKING STOCK

If cost of effort is not too high, offering high powered incentives benefits the leader

The bonus must be high if the marginal cost of effort is high

The base salary is low: equal to agent's outside option

This contract is **risky**:

- ▶ Hard working agent who gets unlucky receives small compensation
- ▶ Ex post, such an agent may perceive her pay to be unfair
- ▶ But she accepts the contract ex ante

RETENTION RULE VERSUS PERFORMANCE PAY

Retention rule:

- ▶ Agent receives wage w only if her performance is above certain threshold $\bar{y}$
- ▶ Is fired and receives 0 otherwise

In the model above, retention rule is equivalent to performance pay

- ▶ Agent needs to receive $u + \alpha$ in expectation to accept the contract
- ▶ Principal needs to pay $u + \alpha$ in expectation to elicit effort

This equivalence is not always true: we will go back to that later

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INTRINSIC MOTIVATION DECREASES NEED FOR PERFORMANCE PAY

An Agent with intrinsic motivation receives extra intrinsic benefit from better performance

This is equivalent to effort being less costly

Lower bonuses needed to elicit effort

But there is a caveat....

CROWDING OUT INTRINSIC MOTIVATION

Experiment by Gneezy and Rustichini, QJE 2000

Hired students to participate in charitable drive soliciting donations

Divided students into three groups:

- ▶ Group 1: received motivational speech
- ▶ Group 2: received speech plus 1% of donations
- ▶ Group 3: received speech plus 10% of donations

Group 1 performed best; Group 2 performed worst

Monetary bonuses may discourage effort!

CROWDING OUT INTRINSIC MOTIVATION

Gneezy and Rustichini's explanation:

The main conclusions of these studies were that positive rewards, in particular monetary rewards, have a negative effect on intrinsic motivation. If a person is rewarded for performing an interesting activity, his intrinsic motivation decreases. The negative effect is significant only if the reward is contingent on the performance; subjects who are paid a fixed positive amount, independent of their performance, do not display reduction in intrinsic motivation.

WHAT WORKS AND WHAT DOESN'T

Research suggests that performance pay is more likely to crowd out intrinsic motivation if

- ▶ Bonuses send signal that effort is purely transactional (Frey & Oberholzer-Gee, 1997)
- ▶ Performance based pay requires excessive monitoring—undermines autonomy and trust (Falk & Kosfeld, 2006)
- ▶ Luck element is large—loss of agency (Prendergast, 1999)
- ▶ Monetary incentives cause distortions—more on that later

When implementing performance pay details matter!

WHAT TO DO IN PRACTICE?

Rely on intrinsic motivation when the following holds:

- ▶ Strong mission alignment and professional identity present
- ▶ Career concerns align incentives with the leader's goals
- ▶ Moderate effort is needed
- ▶ Monetary incentive schemes are expensive—next lecture

Small pay for volunteer work may backfire

Competitive pay for professional work more likely to work

INCENTIVES INTERACT WITH SELECTION

Selection or right workers is important

- ▶ If the job relies on intrinsic motivation, it is important to select workers with high intrinsic motivation
- ▶ If the job uses performance pay, selection on intrinsic motivation matters less

Incentive schemes may affect who the job attracts

- ▶ Performance pay attracts workers who are not intrinsically motivated
- ▶ It may even dissuade those with strong intrinsic motivation

Incentive schemes may affect retention of current agents

- ▶ Introducing performance pay in job that previously relies on intrinsic motivation can backfire

EXAMPLE: SALESPERSON

Individual sales are observable and contractible

Individual sales are closely tied to effort

Sales of commercial product unlikely to sustain strong intrinsic motivation

Implications:

- ▶ High-powered incentives are common

EXAMPLE: TEACHER

Output (student learning, critical thinking) is difficult to measure

Large luck element: difficult to disentangle teacher's effort from student background and environment, peer effects...

Mission-alignment: teachers often believe in the value of what they teach

Implications:

- ▶ **Fixed wage** contracts
- ▶ Reliance on **intrinsic motivation** and career concerns
- ▶ Teachers often have a strong *esprit de corps* (professional norms)

KEY POINTS

Group members exert effort when

- ▶ They have strong intrinsic motivation
- ▶ Their career concerns align with leader's goals
- ▶ The face performance based pay

Monetary incentives work but

- ▶ Are expensive
- ▶ May crowd out intrinsic motivation

What to do:

- ▶ Tailor the incentive scheme to the situation
- ▶ When offering performance pay, details matter